

Chapter 8: Inferences Based on One Sample

Probability and Statistics for Science and Engineering
with Examples in R

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(1) Inferences Concerning a Mean

(A) Normal population with known σ

(a) CI (Confidence Interval)

Random sample $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, σ is known $\Rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

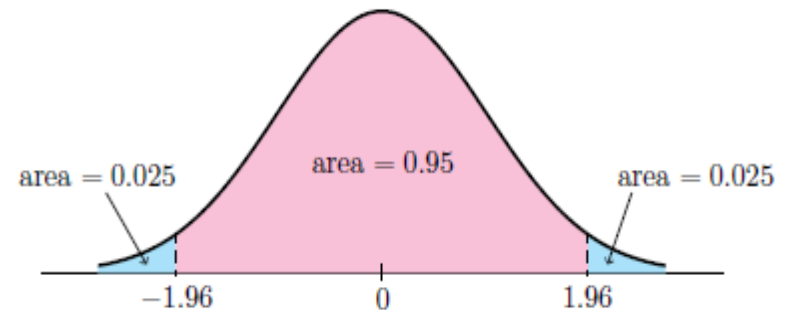
$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$z_{0.025} = 1.96$$

$$P\left(-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right) = .95$$

$$P\left(-1.96 \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < 1.96 \frac{\sigma}{\sqrt{n}}\right) = .95$$

$$P\left(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}\right) = .95$$

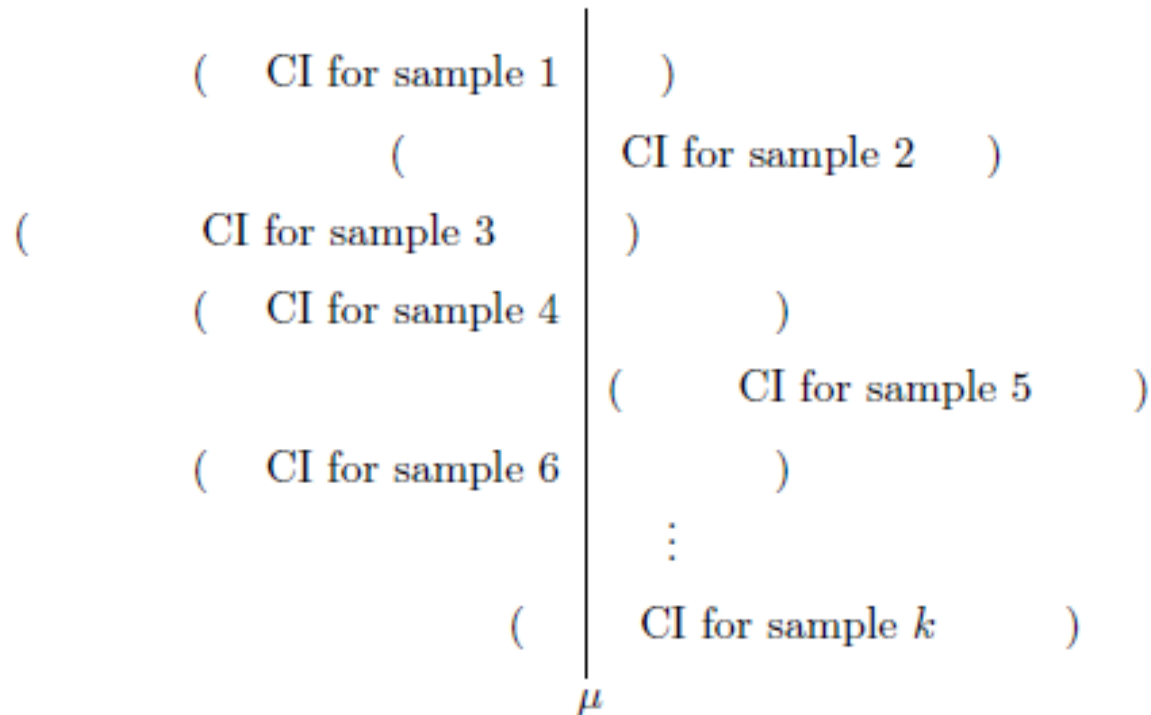


95% CI for μ of $N(\mu, \sigma^2)$ when σ is known:

$$\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}}\right)$$

Interpretation of CIs

If we construct 95% CIs with different samples, then in the long run, 95% of the CIs will contain μ .



A $100(1-\alpha)\%$ CI for μ of $N(\mu, \sigma^2)$ when σ is known:

$$(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}})$$

Values of $z_{\alpha/2}$:

$1-\alpha$	.80	.90	.95	.99
$z_{\alpha/2}$	1.28	1.645	1.96	2.58

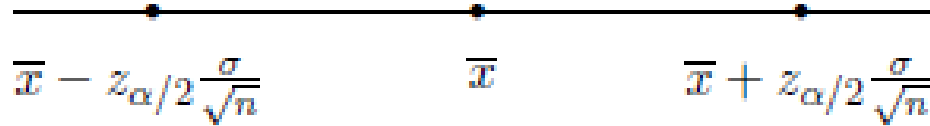
Example 8.1: $X_1, \dots, X_n$: iid from $N(\mu, 4)$, $n = 25$, $\bar{x} = 19.5$.
90% CI for μ ?

$$z_{\alpha/2} = z_{.05} = 1.645$$

$$\bar{x} \pm z_{.05} \frac{\sigma}{\sqrt{n}} = 19.5 \pm 1.645 \frac{2}{\sqrt{25}} = 19.5 \pm 0.66 \rightarrow (18.84, 20.16)$$

(b) Sample size for estimation of μ

With a specific level of significance α , a $100(1-\alpha)\%$ CI can be constructed:



The maximum error of estimate: $E = |\bar{x} - \mu|$

$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$: decreases as n increases

Sample size needed to attain maximum error of estimate E :

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2$$

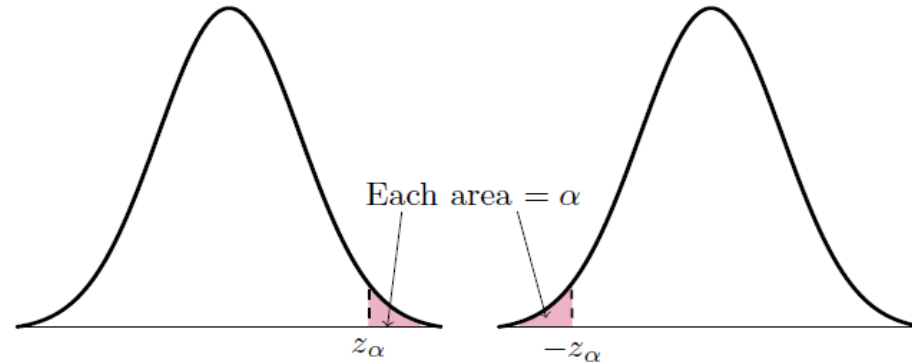
Example 8.2:

A research worker wants to know the average drying time of a certain type of paint under a specific condition, and she wants to be able to assert with 95% confidence that the mean of her sample is off by at most 30 seconds. If $\sigma = 1.5$ minutes, how large a sample will she have to take?

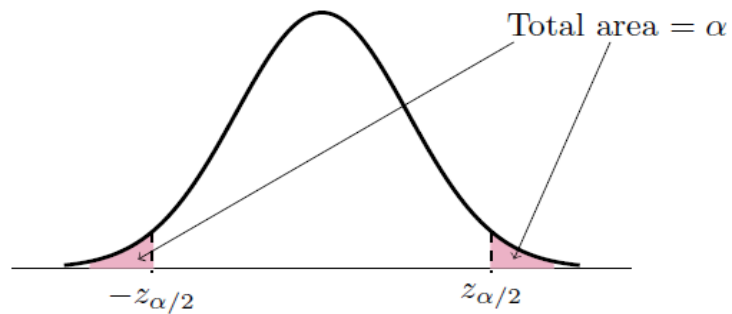
$$n = \left(\frac{Z_{\alpha/2} \sigma}{E} \right)^2 = \left(\frac{Z_{0.025} \sigma}{E} \right)^2 = \left(\frac{1.96 \cdot 1.5}{0.5} \right)^2 = 34.6 \uparrow 35$$

Always round up!

(c) Testing hypothesis



One-sided tests



Two-sided test

Shaded area: rejection region

Test about μ for a normal population with known σ

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \mu = \mu_0, H_1: \mu \neq \mu_0$ (2-sided alternative)	$H_0: \mu = \mu_0, H_1: \mu > \mu_0$ (1-sided alternative)	$H_0: \mu = \mu_0, H_1: \mu < \mu_0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ <u>Rejection Region:</u> $ z \geq z_{\alpha/2}$	$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ <u>Rejection Region:</u> $z \geq z_{\alpha}$	$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ <u>Rejection Region:</u> $z \leq -z_{\alpha}$
Step 4	Substitute $\bar{x}$, σ and n Calculate z Decision	Substitute $\bar{x}$, σ and n Calculate z Decision	Substitute $\bar{x}$, σ and n Calculate z Decision
Step 5	$p = 2P(Z \geq z)$	$p = P(Z \geq z)$	$p = P(Z \leq z)$

Example 8.3: A manufacturer claims that the output for a certain electric circuit is 130 V. A sample of $n = 9$ independent readings on the voltage for this circuit, when tested, yields $\bar{x} = 131.4$ V. It is assumed that the population has a normal distribution with $\sigma = 1.5$ V. Do the data contradict the manufacturer's claim at $\alpha = 0.01$?

$$H_0: \mu = \mu_0 = 130, \quad H_1: \mu \neq 130$$

$$\alpha = .01$$

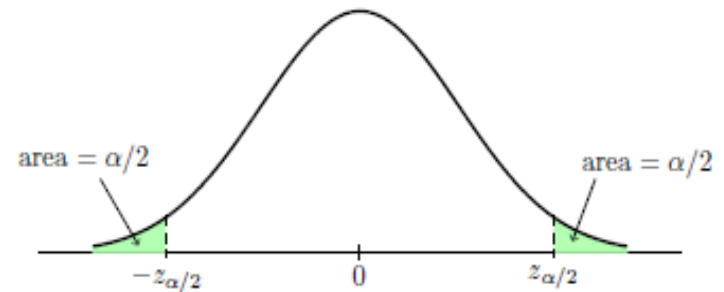
$$\text{Test statistic: } Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

$$\text{Rejection region: } |z| \geq z_{\alpha/2} = z_{.005} = 2.58$$

$$z = \frac{131.4 - 130}{1.5/\sqrt{9}} = 2.8$$

Since $|2.8| = 2.8 > 2.58$, we reject H_0 .

$$p\text{-value} = P(|Z| > 2.8) = 2[1 - \Phi(2.8)] = 2(1 - .9974) = 0.0052 < 0.01$$



Since we rejected H_0 (that the mean output is equal to 130V), yes, the data **do** contradict the manufacturer's claim at level $\alpha = 0.01$.

(d) Relationship between CI & test

- CI = acceptance region in a two-sided test
- The parameter is included in the CI $\Leftrightarrow$ Do not reject H_0 in a two-sided test
- The parameter is not included in the CI $\Leftrightarrow$ Reject H_0 in a two-sided test

Example 8.4: From Example 8.3, a 99% CI for μ :

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 131.4 \pm 2.58 \frac{1.5}{\sqrt{9}} = 131.4 \pm 1.29 \rightarrow (130.11, 132.69)$$

Reject H_0 because $\mu = 130$ is **not** included in the CI.

Probability of a type II error and sample size determination

One sided: the rejection region $Z > z_\alpha$ is equivalent to

$$\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > z_\alpha \text{ and } \bar{X} > \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}$$

$$\beta(\mu') = P(H_0 \text{ is not rejected when } \mu = \mu')$$

$$\begin{aligned} &= P\left(\bar{X} \leq \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} \text{ when } \mu = \mu'\right) = P\left(\frac{\bar{X} - \mu'}{\sigma/\sqrt{n}} \leq \frac{\mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} - \mu'}{\sigma/\sqrt{n}} \middle| \mu = \mu'\right) \\ &= \Phi\left(z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right) = \beta \end{aligned}$$

$$-z_\beta = z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}} \Rightarrow z_\alpha + z_\beta = -(\mu_0 - \mu') \frac{\sqrt{n}}{\sigma} \Rightarrow n = \left(\frac{\sigma(z_\alpha + z_\beta)}{\mu_0 - \mu'}\right)^2$$

Valid for either upper or lower-tailed tests

Probability of a type II error for μ

Alternative Hypothesis	$\beta(\mu')$ for a level α test
$H_1: \mu > \mu_0$	$\Phi\left(z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$
$H_1: \mu < \mu_0$	$1 - \Phi\left(-z_\alpha + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$
$H_1: \mu \neq \mu_0$	$\Phi\left(z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right) - \Phi\left(-z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right)$

Sample size determination for μ

Alternative Hypothesis	Required Sample Size
One-sided	$n = \left(\frac{\sigma(z_\alpha + z_\beta)}{\mu_0 - \mu'} \right)^2$
Two-sided	$n = \left(\frac{\sigma(z_{\alpha/2} + z_\beta)}{\mu_0 - \mu'} \right)^2$

Example 8.5 Let μ denote the true average duration of treating a disease by a certain therapy. Test $H_0: \mu = 20$ days versus $H_1: \mu > 20$ days. $n = 16$ from a normal population distribution with $\sigma = 2$.
 $\alpha = 0.01 \Rightarrow z_\alpha = z_{0.01} = 2.33$

The probability of a type II error when $\mu = 21$ is

$$\beta(21) = \Phi\left(2.33 + \frac{20 - 21}{2/\sqrt{16}}\right) = \Phi(0.33) = 0.6293$$

$$z_{0.1} = 1.28$$

The requirement that the level 0.01 test also has $\beta(21) = 0.1$ is

$$n = \left(\frac{\sigma(z_\alpha + z_\beta)}{\mu_0 - \mu'}\right)^2 = \left(\frac{2(2.33 + 1.28)}{20 - 21}\right)^2 = (-7.22)^2 = 52.13$$

n should be at least 53.

(B) Large Sample with Unknown σ

(a) CI:

Let $X_1, \dots, X_n$ be a random sample from a population with mean μ and variance σ^2 . By CLT,

$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ is approximately $N(0,1)$ if n is large.

$$\therefore P\left(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) \approx 1 - \alpha$$

In most of the cases, σ^2 is unknown.

$Z = \frac{\bar{X} - \mu}{s/\sqrt{n}}$ is approximately $N(0,1)$ if n is large ($n \geq 30$).

A $100(1-\alpha)\%$ CI for μ is $\left(\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}}\right)$.

Example 8.6: To estimate the average speed of cars on a specific highway, an investigator collected speed data from a random sample of 75 cars driving on the highway. $\bar{x} = 58$ mph, $s = 15$ mph. Calculate (1) 90% CI and (2) 80% CI for the mean.

$$\begin{aligned} (1) \text{ 90\% CI: } \alpha/2 = 0.05, z_{\alpha/2} = 1.645, z_{\alpha/2} \frac{s}{\sqrt{n}} &= 1.645 \frac{15}{\sqrt{75}} = 2.85 \\ \left(\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}} \right) &= (58 - 2.85, 58 + 2.85) \\ &= (55.15, 60.85) \end{aligned}$$

$$\begin{aligned} (2) \text{ 80\% CI: } \alpha/2 = 0.1, z_{\alpha/2} = 1.28, z_{\alpha/2} \frac{s}{\sqrt{n}} &= 1.28 \frac{15}{\sqrt{75}} = 2.22 \\ (58 - 2.22, 58 + 2.22) &= (55.78, 60.22) \end{aligned}$$

(b) Testing Hypothesis

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \mu = \mu_0, H_1: \mu \neq \mu_0$ (2-sided alternative)	$H_0: \mu = \mu_0, H_1: \mu > \mu_0$ (1-sided alternative)	$H_0: \mu = \mu_0, H_1: \mu < \mu_0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ <u>Rejection Region:</u> $ z \geq z_{\alpha/2}$	$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ <u>Rejection Region:</u> $z \geq z_{\alpha}$	$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ <u>Rejection Region:</u> $z \leq -z_{\alpha}$
Step 4	Substitute $\bar{x}$, s and n Calculate z Decision	Substitute $\bar{x}$, s and n Calculate z Decision	Substitute $\bar{x}$, s and n Calculate z Decision
Step 5	$p = 2P(Z \geq z)$	$p = P(Z \geq z)$	$p = P(Z \leq z)$

Example 8.7: The duration of treating a disease by a standard therapy has a mean of 15 days. It is claimed that a new therapy can reduce the treatment time. To test this claim, the new therapy is to be tried on 70 patients and their times to recovery are to be recorded.

$\bar{x} = 14.6$ days, $s = 3.0$ days, $n = 70$.

Conduct a hypothesis test using $\alpha = 0.05$.

$$H_0: \mu = 15, H_1: \mu < 15$$

$$\alpha = 0.05$$

$$\text{Test statistic: } Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

$$\text{Rejection region: } Z \leq -z_{\alpha} = -z_{.05} = -1.645$$

$$z = \frac{14.6 - 15}{3/\sqrt{70}} = -1.12 > -1.645$$

Do not reject H_0 . We do not have sufficient evidence to support the claim that the new therapy reduces treatment time.

$$p\text{-value} = P(Z \leq -1.12) = \Phi(-1.12) = 0.1314$$

(C) Small Sample (unknown σ ; normal population)

(a) CI

A $100(1-\alpha)\%$ CI for μ :

$$\left(\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \right) \text{ for } n < 30$$

Example 8.8: Weights of 2-month-old babies in pounds:

8.9, 8.6, 8.0, 8.3, 8.8, 8.6, 8.1, 7.2, 8.0, 8.6, 9.1, 9.0, 9.1, 8.3, 7.9

Assuming normal population, construct a 95% CI for mean.

$$n = 15, \bar{x} = 8.433, s = 0.533$$

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}} = \bar{x} \pm t_{0.025, 15-1} \frac{s}{\sqrt{n}} = 8.433 \pm 2.145 \frac{0.533}{\sqrt{15}} = 8.433 \pm 0.295$$

95% CI for μ : (8.138, 8.728)

(b) Testing hypothesis

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \mu = \mu_0, H_1: \mu \neq \mu_0$ (2-sided alternative)	$H_0: \mu = \mu_0, H_1: \mu > \mu_0$ (1-sided alternative)	$H_0: \mu = \mu_0, H_1: \mu < \mu_0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ <u>Rejection Region:</u> $ t \geq t_{\alpha/2, n-1}$	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ <u>Rejection Region:</u> $t \geq t_{\alpha, n-1}$	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ <u>Rejection Region:</u> $t \leq -t_{\alpha, n-1}$
Step 4	Substitute $\bar{x}$, s and n Calculate t Decision	Substitute $\bar{x}$, s and n Calculate t Decision	Substitute $\bar{x}$, s and n Calculate t Decision
Step 5	$p = 2P(T \geq t)$	$p = P(T \geq t)$	$p = P(T \leq t)$

Example 8.9: IQ's of 5 students from one area of a city:

98, 117, 102, 111, 109

Assuming the IQ score is normally distributed, do the data suggest that the population mean IQ exceeds 100? Use $\alpha = 0.05$.

$$H_0: \mu = 100, H_1: \mu > 100$$

$$\alpha = 0.05$$

$$\text{Test statistic: } T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

$$\text{Rejection region: } t \geq t_{\alpha, n-1} = t_{0.05, 4} = 2.132$$

$$\bar{x} = 107.4, s^2 = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n-1} = 56.3, s = \sqrt{56.3} = 7.5$$

$$t = \frac{107.4 - 100}{7.5/\sqrt{5}} = 2.206 > 2.132 \Rightarrow \text{Reject } H_0$$

$$t_{0.05, 4} = 2.132 < 2.206 < 2.776 = t_{0.025, 4}$$
$$0.025 < p\text{-value} < 0.05$$

Yes, the data suggest that the population mean IQ exceeds 100 at level $\alpha = 0.05$

Finding the p-value using R:

```
>1-pt(2.206, 4)
```

```
0.046176
```

Conducting a test for Example 8.9 using R:

```
>x=c(98,117,102,111,109)
```

```
>t.test(x, mu=100, alt= "greater")
```

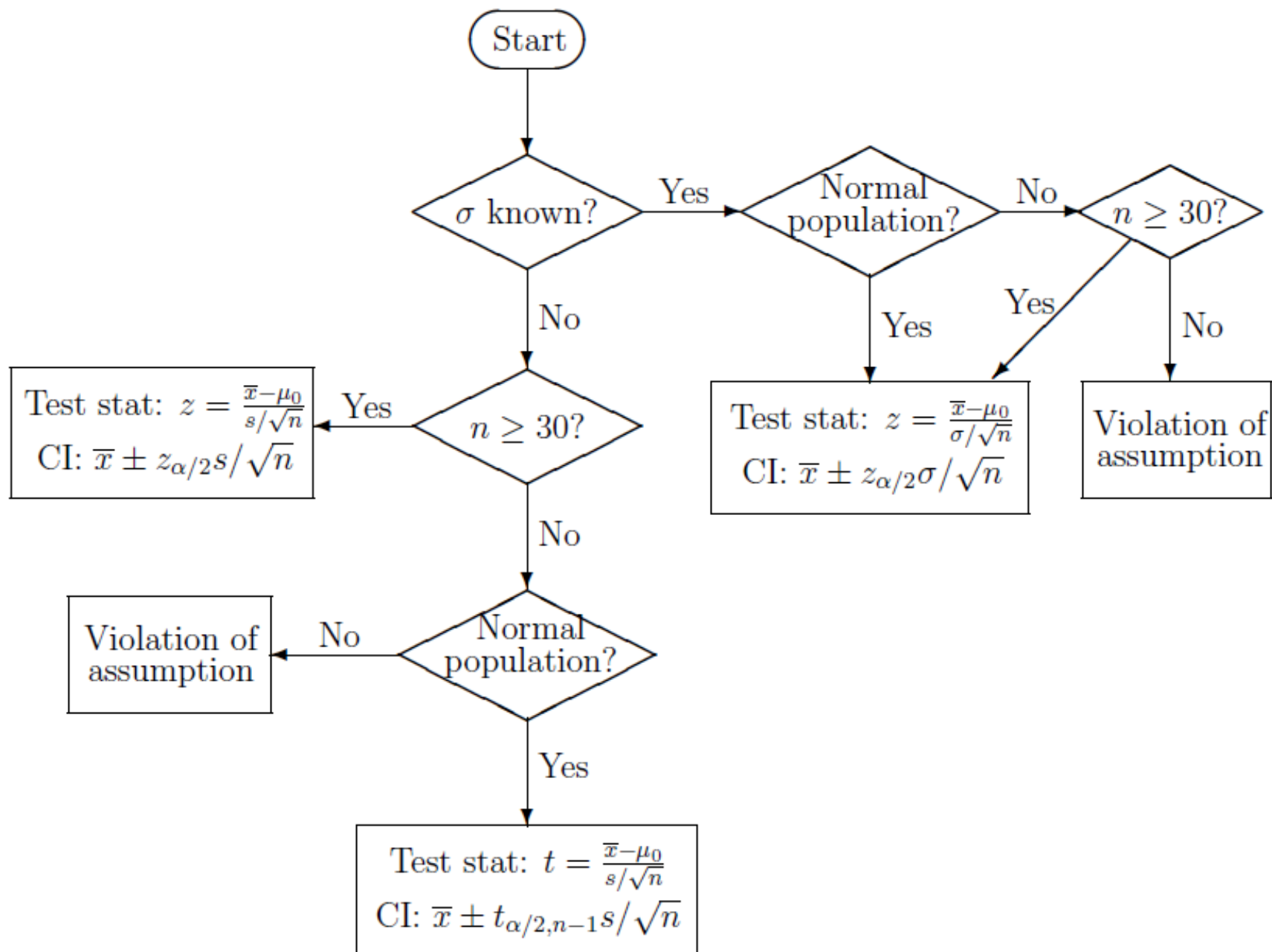
Default significance level is 0.05.

Constructing a CI obtained in Example 8.8 using R:

```
>x=c(8.9,8.6,8.0,8.3,8.8,8.6,8.1,7.2,8.0,8.6,9.1,9.0,9.1,8.3,7.9)
```

```
>t.test(x) # for a 95% CI
```

```
> t.test(x,conf.level=0.90) # for a 90% CI
```



(2) Inferences Concerning a Proportion

(Large sample; for $np \geq 10, n(1 - p) \geq 10$)

X : # successes, p : success rate, n : # trials

$\Rightarrow X \sim \text{Bin}(n, p), E(X) = np$

$\text{Var}(X) = np(1 - p), sd(X) = \sqrt{np(1 - p)}$

$$\hat{p} = \frac{X}{n}$$

$Z = \frac{X - np}{\sqrt{np(1 - p)}}$ is approximately $N(0, 1)$ for large n

$$Z = \frac{X - np}{\sqrt{np(1 - p)}} = \frac{(X - np)/n}{\sqrt{np(1 - p)}/n} = \frac{(X/n) - p}{\sqrt{np(1 - p)}/n^2} = \frac{\hat{p} - p}{\sqrt{p(1 - p)}/n}$$

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = \frac{1}{n} \cdot np = p$$

$$\text{Var}(\hat{p}) = \text{Var}\left(\frac{X}{n}\right) = \frac{1}{n^2} \text{Var}(X) = \frac{1}{n^2} \cdot np(1 - p) = \frac{p(1 - p)}{n}$$

$$sd(\hat{p}) = \sqrt{\frac{p(1 - p)}{n}} \rightarrow \text{Estimate } sd(\hat{p}) \text{ as } \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

(a) Confidence interval

A $100(1 - \alpha)\%$ CI for p for large n :

$$\left(\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}, \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \right)$$

Good for $n\hat{p} \geq 10, n(1 - \hat{p}) \geq 10$

Example 8.10: 820 voters support a certain candidate out of a random sample of 1500 people in a presidential election poll. Find a 95% confidence interval for the support rate of this candidate.

$$\hat{p} = x/n = 820/1500 = 0.547$$

$$z_{\alpha/2} \sqrt{\hat{p}(1 - \hat{p})/n} = 1.96 \sqrt{(0.547)(0.453)/1500} = 0.025$$

$$\begin{aligned} & \left(\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}, \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \right) \\ &= (0.547 - 0.025, 0.547 + 0.025) \\ &= (0.522, 0.572) \end{aligned}$$

Using R,

```
>prop.test(820, 1500, correct=F)
```

(b) Sample size for the estimation of p



The maximum error of estimate: $E = \left| \frac{x}{n} - p \right|$

$$E = z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

Sample size needed to attain maximum error of estimate E :

$$n = p(1-p) \left(\frac{z_{\alpha/2}}{E} \right)^2$$

$p(1-p)$ is maximized when $p = \frac{1}{2} \rightarrow p(1-p) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

$$n = \frac{1}{4} \left(\frac{z_{\alpha/2}}{E} \right)^2 \text{ regardless of the value of } p \text{ (a conservative approach)}$$

Example 8.11: A public health survey is to be designed to estimate the proportion p of a population having a certain disease. How many persons should be examined if the investigators wish to be 99% certain that the error of estimation is below 0.04 when

(a) there is no knowledge about the value of p ?

$$n = \frac{1}{4} \left(\frac{Z_{\alpha/2}}{E} \right)^2 = \frac{1}{4} \left(\frac{Z_{0.005}}{E} \right)^2 = \frac{1}{4} \left(\frac{2.58}{0.04} \right)^2 = 1040.1 \uparrow 1041$$

(b) p is known to be about 0.2?

$$n = p(1 - p) \left(\frac{Z_{\alpha/2}}{E} \right)^2 = (0.2)(0.8) \left(\frac{2.58}{0.04} \right)^2 = 665.64 \uparrow 666$$

(c) Testing hypothesis

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: p = p_0, H_1: p \neq p_0$ (2-sided alternative)	$H_0: p = p_0, H_1: p > p_0$ (1-sided alternative)	$H_0: p = p_0, H_1: p < p_0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$ <u>Rejection Region:</u> $ z \geq z_{\alpha/2}$	$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$ <u>Rejection Region:</u> $z \geq z_{\alpha}$	$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$ <u>Rejection Region:</u> $z \leq -z_{\alpha}$
Step 4	Substitute $\hat{p}$ and n Calculate z Decision	Substitute $\hat{p}$ and n Calculate z Decision	Substitute $\hat{p}$ and n Calculate z Decision
Step 5	$p = 2P(Z \geq z)$	$p = P(Z \geq z)$	$p = P(Z \leq z)$

Example 8.12: A concerned group of citizens want to show that less than a half of the voters support the President's handling of a recent crisis. A random sample of 500 voters gives 228 in support. Does this provide strong evidence for concluding that fewer than a half of all voters support the President's handling of the crisis? Conduct a hypothesis test using $\alpha = 0.01$.

$$\hat{p} = \frac{x}{n} = \frac{228}{500} = 0.456$$

$$H_0: p = 0.5, H_1: p < 0.5$$

$$\alpha = 0.01$$

$$\text{Test statistic: } z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

$$\text{Rejection region: } z < -z_\alpha = -z_{0.01} = -2.33$$

$$z = \frac{0.456 - 0.5}{\sqrt{(0.5)(0.5)/500}} = -1.97 > -2.33 \Rightarrow \text{Do not reject } H_0$$

We do not have sufficient evidence that fewer than a half of the voters support the President's handling of the crisis.

$$p\text{-value} = P(Z < -1.97) = \Phi(-1.97) = 0.024$$

Using R,

```
>prop.test(228, 500, p=0.5, alt= "less", correct=F)
```

Probability of a type II error for p

Alternative Hypothesis	$\beta(p')$ for a Level α Test
$H_1: p > p_0$	$\Phi\left(\frac{p_0 - p' + z_\alpha \sqrt{p_0(1 - p_0)/n}}{\sqrt{p'(1 - p')/n}}\right)$
$H_1: p < p_0$	$1 - \Phi\left(\frac{p_0 - p' - z_\alpha \sqrt{p_0(1 - p_0)/n}}{\sqrt{p'(1 - p')/n}}\right)$
$H_1: p \neq p_0$	$\Phi\left(\frac{p_0 - p' + z_{\alpha/2} \sqrt{p_0(1 - p_0)/n}}{\sqrt{p'(1 - p')/n}}\right) - \Phi\left(\frac{p_0 - p' - z_{\alpha/2} \sqrt{p_0(1 - p_0)/n}}{\sqrt{p'(1 - p')/n}}\right)$

Sample size determination for p

Alternative Hypothesis	Required Sample Size
One-sided	$n = \left(\frac{z_{\alpha} \sqrt{p_0(1 - p_0)} + z_{\beta} \sqrt{p'(1 - p')}}{p' - p_0} \right)^2$
Two-sided	$n = \left(\frac{z_{\alpha/2} \sqrt{p_0(1 - p_0)} + z_{\beta} \sqrt{p'(1 - p')}}{p' - p_0} \right)^2$

Example 8.13: An airline official claims that at least 85% of their flights arrive on time. $H_0: p = 0.85$ versus $H_1: p < 0.85$

(a) If only 75% of the flights arrive on time, how likely is it that a level 0.01 test based on 100 flights will detect such a departure from H_0 ?

With $\alpha = 0.01$, $p_0 = 0.85$ and $p' = 0.75$,

$$\beta(0.75) = 1 - \Phi\left(\frac{0.85 - 0.75 - 2.33\sqrt{\frac{(0.85)(0.15)}{100}}}{\sqrt{\frac{(0.75)(0.25)}{100}}}\right) = 1 - \Phi(0.39) = 0.3483$$
$$1 - \beta(0.75) = 0.6517$$

(b) What should the sample size be in order to ensure that the probability of a type II error at $p = 0.75$ is 0.01?

Using $z_\alpha = z_\beta = 2.33$,

$$n = \left(\frac{2.33\sqrt{(0.85)(0.15)} + 2.33\sqrt{(0.75)(0.25)}}{0.75 - 0.85}\right)^2 = 338.89$$

The sample size needs to be $n = 339$.

(3) Inferences Concerning a Variance

(a) CI for σ^2

Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$. Then

$$P\left(\chi_{1-\frac{\alpha}{2}, n-1}^2 < \frac{(n-1)S^2}{\sigma^2} < \chi_{\frac{\alpha}{2}, n-1}^2\right) = 1 - \alpha$$

A $100(1 - \alpha)\%$ CI for σ^2 of a normal population:

$$\left(\frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}\right)$$

Example 8.14: A random sample of 14 eighth grade boys in Stony Brook, from a normal population gives the following heights in inches: 60.3, 62.7, 67.3, 62.3, 64.9, 71.2, 59.7, 59.6, 65.5, 68.7, 60.5, 64.2, 65.1, 59.5 $\bar{x} = 63.7$, $s^2 = 13.562$. Find a 95% confidence interval for σ^2 .

$$s^2 = 13.562, n = 14$$

$$\chi_{\frac{\alpha}{2}, n-1}^2 = \chi_{0.025, 13}^2 = 24.736, \chi_{1-\frac{\alpha}{2}, n-1}^2 = \chi_{0.975, 13}^2 = 5.009$$

$$\begin{aligned} \left(\frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2} \right) &= \left(\frac{(n-1)s^2}{\chi_{0.025, 13}^2}, \frac{(n-1)s^2}{\chi_{0.975, 13}^2} \right) \\ &= \left(\frac{13(13.562)}{24.736}, \frac{13(13.562)}{5.009} \right) = (7.13, 35.20) \end{aligned}$$

(b) Testing hypothesis

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \sigma^2 = \sigma_0^2,$ $H_1: \sigma^2 \neq \sigma_0^2$ (2-sided alternative)	$H_0: \sigma^2 = \sigma_0^2, H_1: \sigma^2 > \sigma_0^2$ (1-sided alternative)	$H_0: \sigma^2 = \sigma_0^2,$ $H_1: \sigma^2 < \sigma_0^2$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ <u>Rejection Region:</u> $\chi^2 \geq \chi_{\alpha/2, n-1}^2$ or $\chi^2 \leq \chi_{1-\alpha/2, n-1}^2$	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ <u>Rejection Region:</u> $\chi^2 \geq \chi_{\alpha, n-1}^2$	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ <u>Rejection Region:</u> $\chi^2 \leq \chi_{1-\alpha, n-1}^2$
Step 4	Substitute s^2 and n Calculate χ^2 Decision	Substitute s^2 and n Calculate χ^2 Decision	Substitute s^2 and n Calculate χ^2 Decision
Step 5	$p = 2\min \{P(X \geq \chi^2),$ $P(X \leq \chi^2)\}$ where $X \sim \chi_{n-1}^2$	$p = P(X \geq \chi^2)$ where $X \sim \chi_{n-1}^2$	$p = P(X \leq \chi^2)$ where $X \sim \chi_{n-1}^2$

Example 8.15: The length of a certain type of screw is acceptable only if $\sigma \leq 0.6$ mm. Use $\alpha = 0.05$ to test $H_0: \sigma = 0.6$ mm against $H_1: \sigma > 0.6$ mm, if the thicknesses of 16 screws of such a type have a standard deviation of 0.75 mm. Assume normal population.

$$H_0: \sigma = \sigma_0 = 0.6, H_1: \sigma > 0.6$$

$$\alpha = 0.05$$

$$\text{Test statistic: } \chi^2 = \frac{(n-1)S^2}{\sigma_0^2}.$$

$$\text{Rejection region: } \chi^2 \geq \chi_{\alpha, n-1}^2 = \chi_{0.05, 15}^2 = 25.00.$$

$$\chi^2 = \frac{15(0.75)^2}{(0.6)^2} = 23.44 < 25.00 \Rightarrow \text{Do not reject } H_0$$

$$\chi_{0.10, 15}^2 = 22.31 < 23.44 < 25.00 = \chi_{0.05, 15}^2$$

We don't have sufficient evidence that the sd of thickness is larger than 0.6 mm.

$$0.05 < p\text{-value} < 0.10$$

Using R,

```
>1-pchisq(23.44,15)
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0.07523417
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